Year 1 AS Statistics and Mechanics

Chapter 5

Probability

Practice Booklet

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5.1 Calculating probabilities

Practice Book, printed page 31.

1. One letter is chosen at random from the word INTERPOLATION.
Calculate the probability that the letter is:

a. an N

b. a vowel

c. before Q in the alphabet.

Hint

Count the number of successful outcomes in each case.

2. A fair six-sided dice is thrown at the same time as a fair four-sided dice.
Calculate the probability that the product of the scores on the two dice is:

a. 15

b. even

c. greater than 10.

Hint

You can draw a sample space diagram showing all the possible outcomes.

3. E/P The heights, h cm, of a group of volleyball players are recorded in the table.

Height, h (cm)	$150 \leq h < 160$	$160 \leq h < 170$	$170 \leq h < 180$	$180 \leq h < 190$	$190 \leq h < 200$
Frequency	7	22	35	16	9

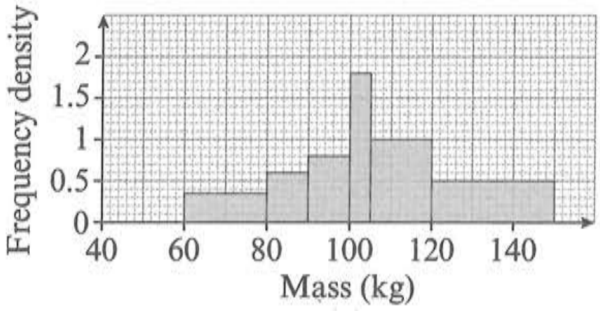
One player is chosen at random.

a. Find the probability that the player is less than 180 cm tall. (2 marks)

b. Estimate the probability that the player is greater than 178 cm tall. (2 marks)

c. State one assumption you have made in calculating your answer to part **b**. (1 mark)

4. E/P The histogram shows the masses, m kg, of 60 adult male warthogs.



One warthog is chosen at random.

a. Find the probability that the warthog has a mass less than 90 kg. (2 marks)

b. Estimate the probability that the warthog has a mass greater than 110 kg. (3 marks)

c. State an assumption you have made in calculating your answer to part **b**. (1 mark)

5. E/P Myra and Nisha have two sets of cards. Set A has the letters B, E, and O printed on them. Set B has the numbers 4, 5, 6 and 7 printed on them. The cards are all face down.
Myra takes one card at random from each set.

a. Write down all of Myra's possible outcomes. (1 mark)

b. Find the probability that Myra selects:

i. the letter B and the number 4

ii. a vowel and an even number.

(2 marks)

Nisha takes two cards, at random, from set B.

c. Write down all of Nisha's possible outcomes. (1 mark)

d. Find the probability that Nisha selects:

i. two odd numbers

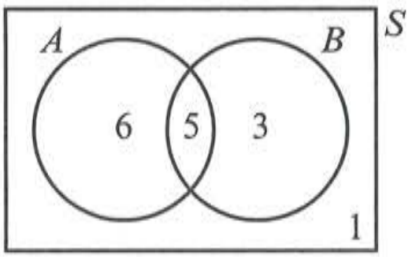
ii. two numbers with a total less than 12.

(2 marks)

5.2 Venn diagrams

Practice Book, printed page 32.

1. Calculate:



a. P(A)
b. P(A and B)
c. P(A or B).

Hint

Calculate the total of all the numbers in the Venn diagram.

2. A group of 20 students are asked if they like tapioca or semolina: 9 students said they liked tapioca, 14 students said they liked semolina, and 5 students said they liked both.
Draw a Venn diagram to illustrate this information.

Hint

Calculate the number of students who liked only tapioca and only semolina.

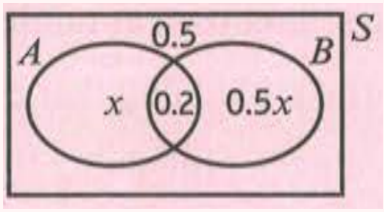
3. Two events A and B have associated probabilities:

$$\begin{aligned} \text{P}(A \text{ and } B) &= 0.2 \\ \text{P}(\text{not } A \text{ and not } B) &= 0.5 \\ \text{P}(A \text{ but not } B) &= 2 \times \text{P}(B \text{ but not } A) \end{aligned}$$

Calculate P(A but not B).

Hint

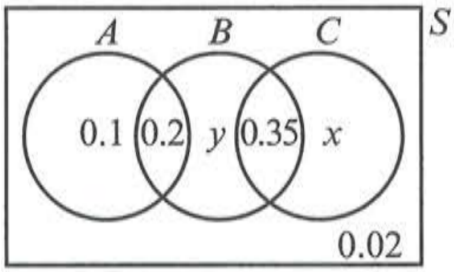
Draw a Venn diagram like this, and use the fact that all the probabilities in the sample space must add up to 1.



4. E/P A passenger on an aeroplane watches a movie with probability 0.8 and a nature documentary with probability 0.3. The probability that the passenger watches either one or both of these options is 0.9.
Find the probability that the passenger watches:

a. both the movie and the nature documentary (2 marks)
b. the nature documentary but not the movie. (2 marks)

5. E/P The Venn diagram shows the probabilities of chickens in a farmyard doing various things in a given period of time.



A represents the event that the chicken pecks.
B represents the event that the chicken clucks.
C represents the event that the chicken lays an egg.
Given that $\mathrm{P}(C) = 0.5$,

a. find x (1 mark)

b. find y. (2 marks)

6. E/P A group of 70 circus performers are asked if they juggle, ride a unicycle or eat fire:

2 performers say that they do all three things
17 performers juggle and ride a unicycle
8 performers ride a unicycle and eat fire
12 performers juggle and eat fire
30 performers juggle
32 performers ride a unicycle
25 performers eat fire

One performer is chosen at random.
Calculate the probability that the performer:

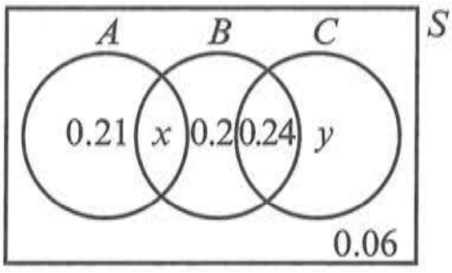
a. juggles (1 mark)

b. juggles or rides a unicycle, or both (2 marks)

c. only eats fire (2 marks)

d. does none of the three things. (2 marks)

7. E/P The Venn diagram shows the probabilities of members of an athletics club taking part in various events.

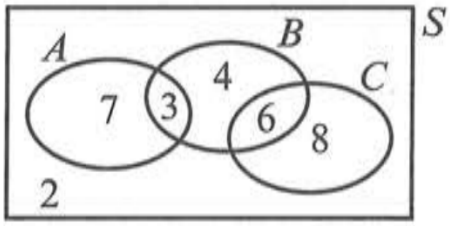


A represents the event that the member takes part in the javelin.
B represents the event that the member takes part in the discus.
C represents the event that the member takes part in the shot put.

a. Given that $\mathrm{P}(B) = 0.62$, find the values of x and y. (3 marks)

b. Calculate the probability that a member chosen at random takes part in either the javelin or the shot put, but not the discus. (2 marks)

8. E/P The Venn diagram shows the number of children at a party who like different types of pudding.



A represents the event that the child likes jelly.
B represents the event that the child likes ice cream.
C represents the event that the child likes blancmange.
One of the children is chosen at random.

a. Show that the probability that the child likes more than one type of pudding is $\frac{3}{10}$ (2 marks)

b. Find the probability that the child likes jelly, or ice cream, or both. (2 marks)

c. Write down the probability that the child likes both ice cream and blancmange. (1 mark)

5.3 Mutually exclusive and independent events

Practice Book, printed page 34.

1. Events A and B are mutually exclusive, and $\mathrm{P}(A) = 0.3$ and $\mathrm{P}(B) = 0.45$.

a. Find $\mathrm{P}(A \text{ or } B)$.

b. Find $\mathrm{P}(\text{neither } A \text{ nor } B)$.

Hint

Draw a Venn diagram.

2. Events A and B are mutually exclusive, and $\mathrm{P}(A \text{ or } B) = 0.7$ and $\mathrm{P}(B) = 0.5$.

a. Find $\mathrm{P}(A)$.

b. Find $\mathrm{P}(\text{neither } A \text{ nor } B)$.

Hint

If two events are mutually exclusive, $\mathrm{P}(A \text{ and } B) = 0$.

3. $\mathrm{P}(A) = 0.6$ and $\mathrm{P}(B) = 0.4$.
Given that events A and B are independent, find $\mathrm{P}(A \text{ and } B)$.

Hint

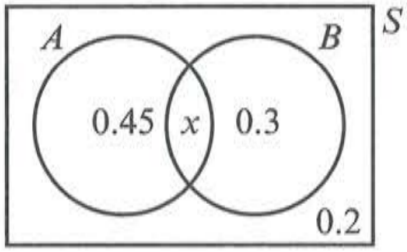
If two events are independent, you can multiply their probabilities to find the probability of both events happening.

4. $\mathrm{P}(A) = 0.12$ and $\mathrm{P}(A \text{ and } B) = 0.0204$.
Given that events A and B are independent, find $\mathrm{P}(B)$.

Hint

You can rearrange the formula for the multiplication rule for independent events.

5. E/P The Venn diagram shows the probabilities that a cat likes *Lixilike* cat food (A) or *Purrfect* cat food (B).



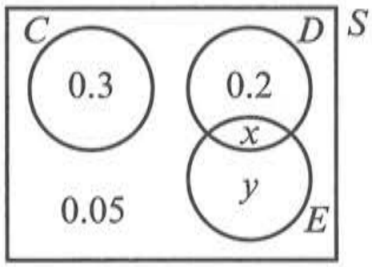
a. Write down the value of x. (1 mark)

b. Determine whether the events 'likes *Lixilike*' and 'likes *Purrfect*' are independent. (3 marks)

6. E/P Y and Z are two events such that $\mathrm{P}(Y) = 0.4$, $\mathrm{P}(Y \text{ and not } Z) = 0.15$ and $\mathrm{P}(\text{neither } Y \text{ nor } Z) = 0.35$.

Show that events Y and Z are not independent. (3 marks)

7. E/P The Venn diagram shows the probabilities of a group of students participating in various activities.



C represents taking part in school council.
D represents taking part in the school drama production.
E represents taking part in the Duke of Edinburgh scheme.
The probability that a student takes part in school council or the drama production is 0.65.

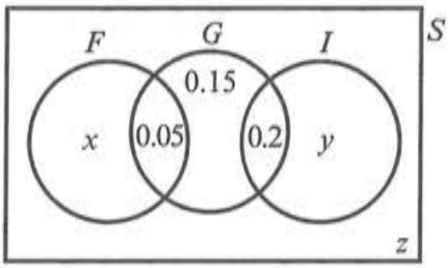
a. Write down two events that are mutually exclusive. (1 mark)

b. Find the value of x and the value of y. (2 marks)

A student claims that events D and E are independent.

c. Assess the student's claim. (3 marks)

8. E/P Frank, Gracie and Igor sometimes miss their train to work.
F represents the event that Frank misses his train.
G represents the event that Gracie misses her train.
I represents the event that Igor misses his train.



The Venn diagram shows the three events, F, G and I and the probabilities associated with each region of event G. The constants x, y and z each represent probabilities associated with the three separate regions outside G.

a. Write down two of the events that are mutually exclusive. Give a reason for your answer. (1 mark)

The probability that Frank misses his train is 0.3.
The events G and I are independent.

b. Find the values of x, y and z. (4 marks)

5.4 Tree diagrams

Practice Book, printed page 35.

1. A bag contains four red counters and five yellow counters. A counter is chosen at random, the colour is recorded and the counter is replaced.
A second counter is then chosen at random and the colour is recorded.

a. Draw a tree diagram to show all the possible outcomes from this experiment.

b. Calculate the probability that:

i. both counters are red

ii. only one counter is red.

Hint

Multiply probabilities along each branch and add up the probabilities for all the successful outcomes.

2. In a raffle game, a cup contains 15 tickets numbered from 1 to 15. A player gets to draw two tickets from the cup at random. The player wins a prize for each ticket that is a multiple of 5.

a. Draw a tree diagram to show all the possible outcomes of both draws.

b. Calculate the probability that:

i. the player chooses two winning tickets

ii. the player chooses exactly one winning ticket.

Hint

The first ticket is not replaced, so the probabilities will change for the second draw.

3. E/P The probability that a coach driver takes the motorway is 0.4. If the coach driver takes the motorway, the probability that the coach arrives on time at its destination is 0.9. If the driver does not take the motorway, the probability that the coach arrives on time at its destination is 0.3.

a. Draw a tree diagram to represent this information. (3 marks)

b. State, with a reason, whether the events 'takes the motorway' and 'arrives on time' are independent. (1 mark)

c. Calculate the probability that the coach does not arrive on time at its destination. (3 marks)

4. E/P In an experiment, a biased dice is rolled three times and it is recorded whether it lands on an odd number or not. P(odd) = 0.6.
By drawing a tree diagram, or otherwise, calculate the probability that:

a. all three rolls are odd (2 marks)

b. just one of the three rolls is odd. (3 marks)

The experiment is repeated twice more.

c. Find the probability of obtaining either 3 odd numbers or 3 even numbers all three times. (3 marks)

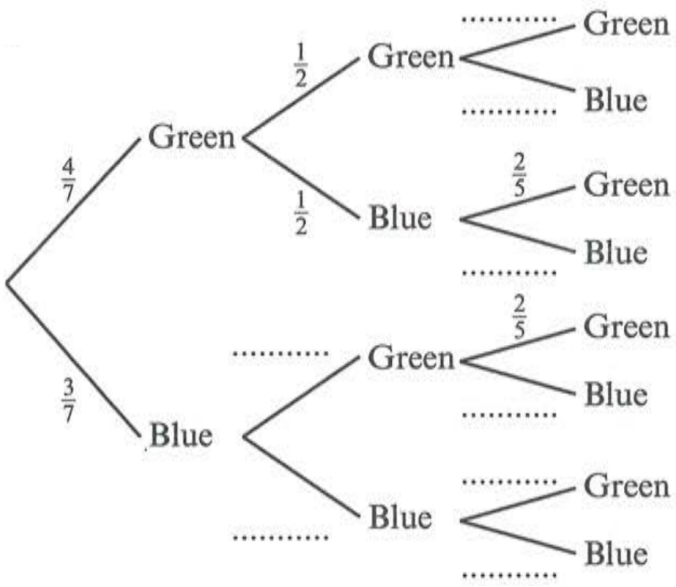
5. E/P A group of patients with a particular disease took part in a study to test three different medications. In the study, 20% of the patients were given medication A, 50% of the patients were given medication B, and the rest of the patients were given medication C.
The researchers found that 6% of the patients in the study were not cured. Given that 5% of the patients who received medication A were not cured and 7% of the patients who received medication B were not cured,

a. find the percentage of patients who received medication C who were not cured. (3 marks)

A patient is chosen at random.

b. Explain why the event 'the patient was given medication B' and the event 'the patient was not cured' are not statistically independent. (1 mark)

6. E/P Two bags contain coloured balls. Bag A contains 7 balls, of which 4 are green and 3 are blue. Bag B contains 8 balls, of which 3 are green and 5 are blue.
A ball is drawn at random from bag A and placed into bag B. A second ball is then drawn at random from bag A and placed into bag B.
A third ball is then drawn at random from the 10 balls in bag B.
The event Q occurs when the two balls drawn from bag A are the same colour.
The event R occurs when the ball drawn from bag B is blue.



a. Copy and complete the tree diagram shown above. (4 marks)

b. Find P(Q). (2 marks)

c. Show that $\mathrm{P}(R) = \dfrac{41}{70}$ (3 marks)

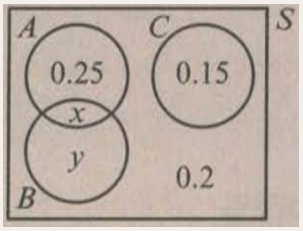
d. Hence write down the value of P(Q and R). (1 mark)

e. State, with a reason, whether events Q and R are independent. (1 mark)

Problem solving: Set A

Bronze

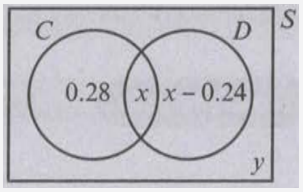
A group of students are interviewed to find the types of food they like.
Event A is 'likes salad', event B is 'likes pasta' and event C is 'likes curry'.
One of these students is chosen at random, and a Venn diagram is drawn to show the probabilities of each outcome.



a. Given that $P(A) = 0.55$, find the values of x and y. (2 marks)
b. Show that events A and B are not statistically independent. (2 marks)

Silver

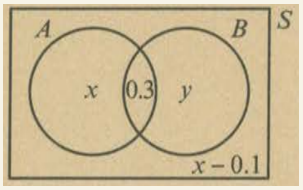
The Venn diagram shows the probabilities of two events, C and D.



a. Given that events C and D are independent show that $2x^2 - 0.68x - 0.0672 = 0$ (3 marks)
b. Solve this quadratic equation, and hence find the values of x and y. (3 marks)
c. Explain briefly why only one solution to the quadratic equation in part **a** should be considered. (1 mark)

Gold

The Venn diagram shows the probabilities of two events, A and B.
Given that events A and B are independent, find the values of x and y. (4 marks)



Problem solving: Set B

Bronze

A bag contains 6 orange tokens and 3 green tokens. A token is drawn out of the bag at random, the colour noted, and then replaced. A second token is then drawn out and the colour noted.

a. Draw a tree diagram to illustrate this situation. (3 marks)
b. Calculate the probability that one token of each colour is drawn. (2 marks)

Silver

A box contains 6 red counters, 3 green counters and 2 yellow counters. A counter is drawn out of the box at random. The colour is recorded and the counter is not replaced. A second counter is then drawn out and the colour is recorded.

a. Draw a tree diagram to illustrate this situation. (3 marks)
b. Calculate the probability that the counters are the same colour. (2 marks)

Gold

A box contains 8 ginger biscuits and 7 oat biscuits. Mathew chooses a biscuit at random. If it is a ginger biscuit, he puts it back. If it is an oat biscuit, he eats it. In either case, he then chooses a second biscuit. Again, if it is a ginger biscuit, he puts it back and if it is an oat biscuit, he eats it. He makes a third selection, with the same outcomes.
Find the probability that Mathew eats exactly two oat biscuits. (4 marks)

Now try this → Exam question bank Q5, Q10, Q25, Q32, Q35

Selected exam question bank questions

Question 5

The daily mean temperatures, $t\,°\text{C}$, for a random sample of 20 days in Hurn during 2015 are taken from the large data set.

Temp, $t\,°\text{C}$	$8 \leq t < 10$	$10 \leq t < 12$	$12 \leq t < 14$	$14 \leq t < 16$	$16 \leq t < 18$
Frequency	2	7	4	5	2

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Two days from the sample are chosen at random.

a. Find the probability that, on one of the two days, the temperature was between 12°C and 14°C. (3 marks)

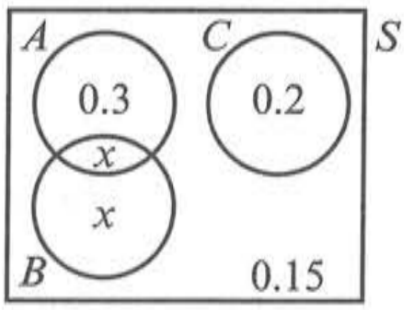
b. State one assumption you have made in calculating your answer to **a**. (1 mark)

Question 10

Some students are interviewed to find the types of music they like.

Event A is 'likes rock music', event B is 'likes pop music' and event C is 'likes classical music'.

A Venn diagram is drawn to show the probabilities in this case.



a. Write down a pair of events that are mutually exclusive. (1 mark)

b. Write down the value of x. (1 mark)

c. Show that events A and B are not statistically independent. (2 marks)

Question 25

A supermarket has three different suppliers for its oranges. Supplier X supplies 20% of the oranges and it is found that 3% of these oranges are spoilt on arrival. Supplier Y supplies 40% of the oranges and it is found that 4% of these oranges are spoilt on arrival. The remaining oranges are supplied by supplier Z.

Overall, 5% of the oranges received by the supermarket are spoilt on arrival.

a. Find the percentage of oranges from supplier Z that are spoilt on arrival. (3 marks)

b. Show that the event 'orange is from supplier X' and the event 'orange is spoilt on arrival' are not statistically independent. (4 marks)

Question 32

The daily mean windspeeds, w knots, for a random sample of 30 days in Camborne in 2015 are taken from the large data set.

Wind speed, w (kn)	$5 \leq w < 8$	$8 \leq w < 11$	$11 \leq w < 14$	$14 \leq w < 17$	$17 \leq w < 20$
Frequency	7	5	6	8	4

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a. Calculate estimates for the mean and standard deviation for this data. (2 marks)

b. Calculate an estimate for the probability that the daily mean windspeed on a day chosen from the sample at random was greater than one standard deviation from the mean. (3 marks)

c. State one assumption that you have made in calculating your answer to part **b**. (1 mark)

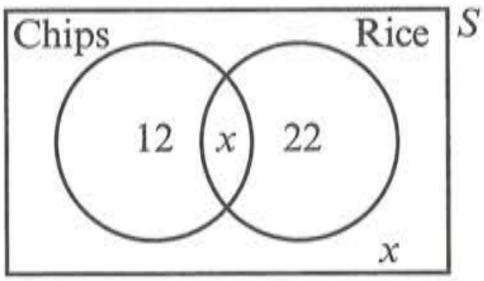
Question 35

Genevieve took a stratified random sample of size 50 from the students in her school. She used the different year groups as her strata.

a. Explain briefly how Genevieve took her sample. (2 marks)

She asked each of the students in the sample if they liked chips or rice.

She summarised her findings in a Venn diagram.



b. Write down the value of x. (1 mark)

One student from the sample is chosen at random.

c. Calculate the probability that they like either chips or rice, or both. (2 marks)

d. Are the events 'like chips' and 'like rice' independent? Justify your answer. (3 marks)

Answers

5.1 Calculating probabilities

1.

a. $\frac{2}{13}$
b. $\frac{6}{13}$
c. $\frac{10}{13}$

2.

a. $\frac{1}{24}$
b. $\frac{3}{4}$
c. $\frac{1}{3}$

3.

a. $\frac{64}{89}$
b. $\frac{32}{89}$
c. The heights are evenly distributed within each class.

4.

a. $\frac{13}{60}$
b. $\frac{5}{12}$
c. The masses are evenly distributed within each class.

5.

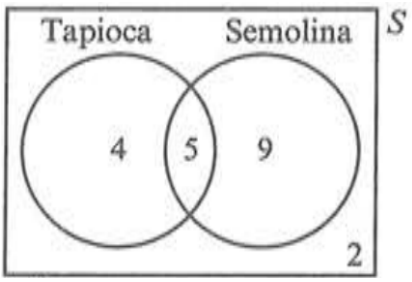
a. B4, B5, B6, B7, E4, E5, E6, E7, O4, O5, O6, O7
b.
i. $\frac{1}{12}$
ii. $\frac{1}{3}$
c. 45, 46, 47, 54, 56, 57, 64, 65, 67, 74, 75, 76
d.
i. $\frac{1}{6}$
ii. $\frac{2}{3}$

5.2 Venn diagrams

1.

a. $\frac{11}{15}$
b. $\frac{1}{3}$
c. $\frac{14}{15}$

2.



3. 0.2

4.

a. 0.2
b. 0.1

5.

a. 0.15
b. 0.18

6.

a. $\frac{3}{7}$
b. $\frac{9}{14}$
c. $\frac{1}{10}$
d. $\frac{9}{35}$

7.

a. $x = 0.18$, $y = 0.11$
b. 0.32

8.

a. $3 + 6 = 9$; $\frac{9}{30} = \frac{3}{10}$
b. $\frac{2}{3}$
c. $\frac{1}{5}$

5.3 Mutually exclusive and independent events

1.

a. 0.75
b. 0.25

2.

a. 0.2
b. 0.3

3. 0.24

4. 0.17

5.

a. 0.05
b. No: $0.5 \times 0.35 \neq 0.05$

6. $\mathrm{P}(Y) = 0.4$; $\mathrm{P}(Z) = 0.5$; $\mathrm{P}(Y) \times \mathrm{P}(Z) = 0.4 \times 0.5 \neq 0.25$

7.

a. C and D or C and E

b. $x = 0.15$; $y = 0.3$

c. $\mathrm{P}(D) \times \mathrm{P}(E) = 0.35 \times 0.45 \neq 0.15$ therefore the student is not correct.

8.

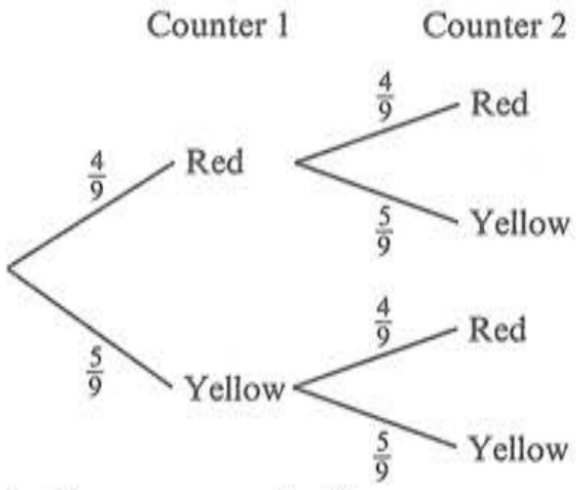
a. F and I, since their circles do not overlap.

b. $x = 0.25$; $y = 0.3$; $z = 0.05$

5.4 Tree diagrams

1.

a.



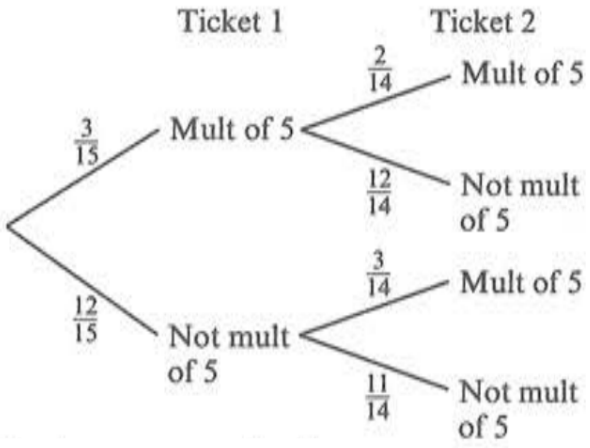
b.

i. $\frac{16}{81}$

ii. $\frac{40}{81}$

2.

a.



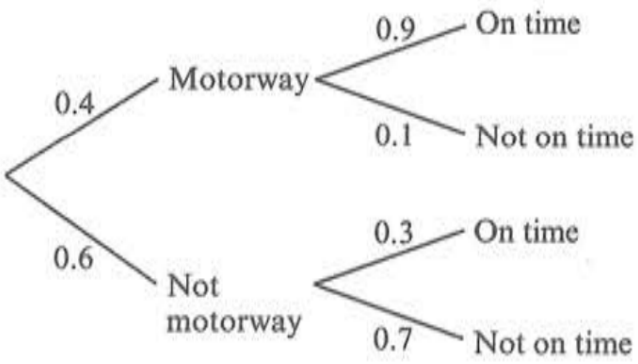
b.

i. $\frac{1}{35}$

ii. $\frac{12}{35}$

3.

a.



b. No: P(on time) is different for each of the two initial choices.

c. 0.46

4.

a. 0.216

b. 0.288

c. $\frac{343}{15625}$

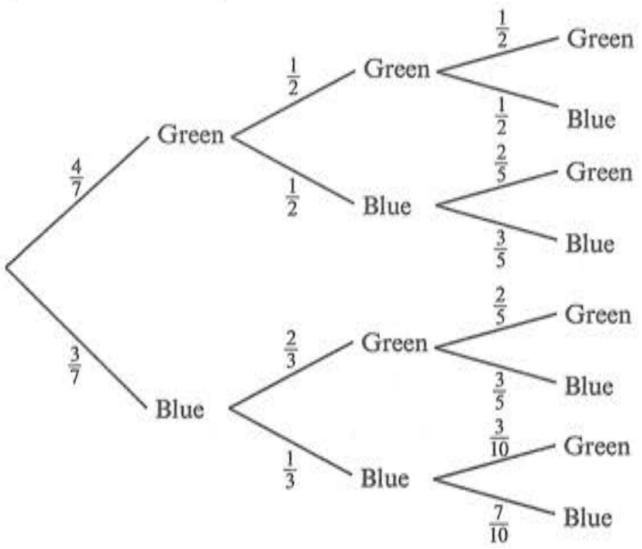
5.

a. 5%

b. P(B and not cured) $= 0.035 \neq$ P(B) $\times$ P(not cured) $(= 0.5 \times 0.06 = 0.03)$

6.

a.



b. $\frac{3}{7}$

c.
$$\begin{aligned} \mathrm{P}(R) &= \frac{4}{7} \times \frac{1}{2} \times \frac{1}{2} + \frac{4}{7} \times \frac{1}{2} \times \frac{3}{5} + \frac{3}{7} \times \frac{2}{3} \times \frac{3}{5} + \frac{3}{7} \times \frac{1}{3} \times \frac{7}{10} \\ &= \frac{41}{70} \end{aligned}$$

d. $\frac{17}{70}$

e. No: P(Q) $\times$ P(R) $\neq$ P(Q and R)

Problem solving: Set A

Bronze

a. $x = 0.3$, $y = 0.1$

b. $\mathrm{P}(A) \times \mathrm{P}(B) = 0.55 \times 0.4 \neq \mathrm{P}(A \text{ and } B)$ $(= 0.3)$

Silver

a. $(0.28 + x)(x - 0.24 + x) = x$ (because of independence)

$$2x^2 + 0.32x - 0.0672 = x$$

$$2x^2 - 0.68x - 0.0672 = 0$$

b. $x = 0.42$, $y = 0.12$

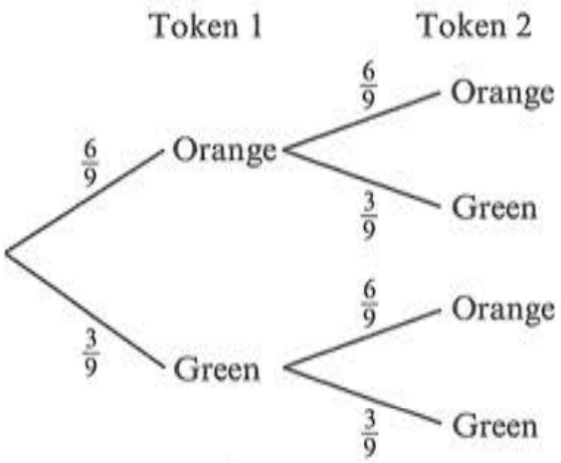
c. Other solution ($x = -0.08$) would give negative probabilities, so should be rejected.

Gold $x = 0.3$, $y = 0.2$

Problem solving: Set B

Bronze

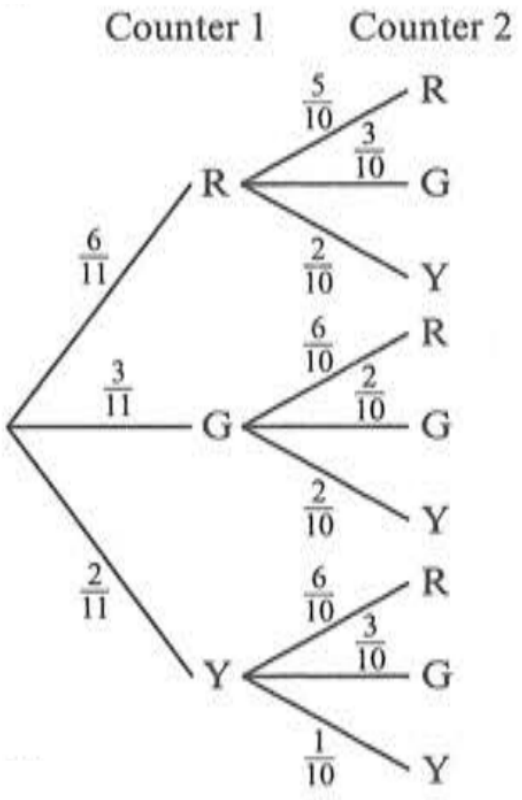
a.



b. $\frac{4}{9}$

Silver

a.



b. $\frac{19}{55}$

Gold $\frac{2348}{6825}$

Selected exam question bank answers

Question 5

a. $\frac{32}{95}$

b. That the same day cannot be chosen twice (non-replacement).

Question 10

a. A and C or B and C

b. 0.175

c. $\mathrm{P}(A) \times \mathrm{P}(B) = 0.475 \times 0.35 \neq \mathrm{P}(A \text{ and } B)$ (which is 0.175)

Question 25

a. 7%

b. $\mathrm{P}(X) \times \mathrm{P}(\text{spoilt}) = 0.2 \times 0.05 \neq \mathrm{P}(X \text{ and spoilt})$ (which is 0.006)

Question 32

a. 12.2; 4.12 (3 s.f.)

b. $\frac{13}{30}$

c. That the data is evenly distributed within each class.

Question 35

a. She chose randomly from the students in each year group in proportion to the number of students in that year group.

b. 8

c. 0.84

d. $0.4 \times 0.6 = 0.24 \neq 0.16$ so not independent.